A FINITE UNION OF NOETHERIAN SUBSETS IS NOETHERIAN FOR THE INDUCED TOPOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Noetherian property of a topological space as the ascending-chain condition for open subsets (equivalently the descending-chain condition for closed subsets), and the induced topology on a subset. Set up in the author's section on Noetherian topological spaces, immediately before the counted statement. Fixes the property to be proved and the topology on the union.

Lemma 0.1. *Let X be a topological space. Let $X_i \subset X$, $i = 1, \ldots, n$ be a finite collection of subsets. If each X_i is Noetherian (with the induced topology), then $\bigcup_{i=1,\ldots,n} X_i$ is Noetherian (with the induced topology).*

Proof. Let $\{F_m\}_{m \in \mathbf{N}}$ a decreasing sequence of closed subsets of $X' = \bigcup_{i=1,\ldots,n} X_i$ with the induced topology. Then we can find a decreasing sequence $\{G_m\}_{m \in \mathbf{N}}$ of closed subsets of X verifying $F_m = G_m \cap X'$ for all m (small detail omitted). As X_i is noetherian and $\{G_m \cap X_i\}_{m \in \mathbf{N}}$ a decreasing sequence of closed subsets of X_i, there exists $m_i \in \mathbf{N}$ such that for all $m \geq m_i$ we have $G_m \cap X_i = G_{m_i} \cap X_i$. Let $m_0 = \max_{i=1,\ldots,n} m_i$. Then clearly

$$F_m = G_m \cap X' = G_m \cap (X_1 \cup \ldots \cup X_n) = (G_m \cap X_1) \cup \ldots (G_m \cap X_n)$$

stabilizes for $m \geq m_0$ and the proof is complete. □

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